

****Predicted Tip using Multiple Linear Regression****

Multiple Linear Regression is a statistical technique used to predict the value of a dependent variable based on two or more independent variables. In this project, the dependent variable is ****Tip****, and the independent variables include ****Total bill, Size, Sex, Smoker, Day, and Time****.

The regression model estimates the relationship between the tip amount and these variables. The general form of the regression equation is:

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + b_4X_4$$

Where:

$\hat{Y}$ = Predicted value of Tip

b_0 = Intercept (constant term)

b_1, b_2, b_3, b_4 = Regression coefficients

X_1, X_2, X_3, X_4 = Independent variables

For this dataset:

X_1 = Total bill

X_2 = Size

X_3 = Day

X_4 = Time

Using this equation, the model predicts the expected tip amount based on the values of the independent variables. The predicted tip helps in understanding how factors such as the total bill amount, number of people at the table, day of the week, and meal time influence tipping behavior in restaurants.